

YEN-LIN (ALLEN) LU

New York, NY | 765-771-0363 | allen930526@gmail.com | <https://www.linkedin.com/in/allen-lu526/> | 1allenlu.github.io/

EDUCATION

Cornell University, Cornell Tech <i>M.Eng. in Computer Science</i>	Aug 2026 – May 2027 <i>GPA: In Progress</i>
New York University Tandon School of Engineering <i>B.S. in Mathematics & Computer Science (Double Major)</i>	Sep 2023 – May 2026 <i>GPA: 3.73/4.00</i>
University of California, Irvine <i>Coursework toward B.S. in Mathematics</i>	Sep 2022 – Jun 2023 <i>GPA: 3.99/4.00</i>

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C, SQL, Swift
Backend & Frameworks: React, Node.js, Express.js, SQLAlchemy, Mongooose
ML & Data: PyTorch, Scikit-learn, XGBoost, SHAP, Pandas, NumPy
Databases & Infra: PostgreSQL, TimescaleDB, Redis, MongoDB, SQLite, MySQL, Docker, Nginx, AWS S3

EXPERIENCE

Research Assistant <i>NYU Cognition and Computation Lab</i>	Oct 2025 – May 2026 <i>New York, NY</i>
- Built a hybrid LLM-Bayesian learning system translating natural language hints into symbolic meaning functions, combining neural language understanding with probabilistic inference for explainable cognitive modeling. (Python, PyTorch, Transformers) - Designed in-context learning pipelines benchmarking LLM-generated against hand-coded meaning functions across 400 simulated episodes, finding 53% vs. 94% decision accuracy respectively. (Python, LLM APIs)	
Student Product Support Analyst <i>New York University</i>	Oct 2024 – May 2026 <i>New York, NY</i>
- Tested and triaged pre-deployment updates to the Brightspace LMS platform, identifying regressions and stability issues affecting 80,000+ users before rollout. (QA Testing, Regression Testing) - Authored technical documentation and delivered 50+ live training sessions, reducing recurring support tickets	
Software Development Engineer Intern <i>Athleo.ai</i>	Jun 2024 – Aug 2024 <i>Boston, MA (Remote)</i>
- Built and shipped a production-facing website with responsive layouts and cross-browser support, sustaining 1,500+ daily active users. (React, JavaScript, HTML/CSS) - Resolved 40+ client-reported bugs through structured QA testing and debugging, improving product reliability and reducing repeat issue reports.	

PROJECTS

QuantStream	Sep 2025 – Jun 2026
- Engineered a full-stack algorithmic trading platform with concurrent, non-blocking pipelines for real-time ML signal generation and execution. (Python, PyTorch, TypeScript) - Reduced signal computation p50 latency from 23ms to 1.2ms (19x) with a 90% cache hit rate by adding a Redis caching layer, validated across a 100+ test suite. (Redis, PostgreSQL, Docker, Pytest)	
RedBusTracker	Dec 2025 – May 2026
- Built a real-time iOS app polling live vehicle and ETA data to track the Roosevelt Island shuttle and Tramway schedule. (Swift, SwiftUI, URLSession) - Implemented a Dynamic Island and Lock Screen Live Activity for background ETA updates, backed by unit tests on parsing and calculation logic. (UIKit, XCTest)	
DBook	Dec 2025 – May 2026
- Architected a full-stack book-tracking platform with query-level authorization scoping, ensuring every user's bookshelf and review data remained isolated even under concurrent multi-user access. (React, Node.js, Express.js, MongoDB) - Integrated an LLM into two recommendation features using structured JSON-mode prompting grounded in live user data, backed by a polymorphic admin schema validated through 74 integration tests. (LLM APIs, Jest)	
EcoBite	Dec 2025 – May 2026
- Built a full-stack sustainability platform that ranks NYC neighborhoods by environmental impact, fusing disparate energy-disclosure and health-inspection datasets into a single comparable score. (React, SQLite, SQLAlchemy) - Designed a computer vision pipeline that turns a photo of any restaurant menu into per-dish carbon estimates in real time, paired with an interactive map for exploring results by neighborhood. (OpenAI Vision, Mapbox GL)	

LEADERSHIP & INVOLVEMENT

Taiwan Armed Forces – Mandatory Military Service (Summer 2025, Summer 2026) | **Applied Math Club** – Co-Founder, Secretary, Treasurer | **NYU Theta Tau Professional Engineering Fraternity** – Webmaster, Treasurer | **NYU VIP RoboMaster Team** – Software Team Member